

E-Commerce Sales & Profitability Analysis

SIC – AI801&803

Agenda

- 01 — Project Overview
- 02 — Business Problem & Objectives
- 03 — Dataset & Data Preparation
- 04 — Exploratory Data Analysis
- 05 — Key Findings & Insights
- 06 — Business Recommendations
- 07 — Conclusion & Future Improvements

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Project Overview

Project Overview

Main Content

E-Commerce Sales & Profitability Analysis

- This project analyzes e-commerce transaction data to understand sales performance, profitability, product performance, geographic trends, payment methods, and customer purchasing behavior.

Key Focus Areas

Sales Performance

- Understanding revenue trends and overall business performance.

Profitability

- Identifying profitable and loss-making products, categories, and regions.

Customer Behavior

- Exploring purchasing patterns and repeat customer activity.

Payment Performance

- Comparing sales, profit, margins, and order value across payment methods.

Goal

Transform e-commerce transaction data into actionable business insights that support better sales and profitability decisions



Business Problem & Objectives

Business Problem

Sales Performance Does Not Always Translate into Profitability

- The e-commerce business generates strong sales activity, but revenue growth does not consistently translate into healthy profitability.

Where Are the Profitability Gaps?

- Some products and sub-categories generate substantial sales while delivering low or negative profit, while certain regions and time periods also show weak profitability.

Why It Matters

- Focusing only on sales volume can hide underperforming areas and lead to decisions that increase revenue without improving the bottom line.

Business Need

- The business needs a clear view of where revenue is being generated, where profit is being lost, and which areas require management attention.



Business Objectives

01 — Measure Overall Performance

Assess total sales, profit, order volume, average order value, and overall profit margin.

02 — Identify Profit Drivers

Determine which categories and sub-categories contribute most to sales and profitability, and identify loss-making areas.

03 — Evaluate Performance by Time & Region

Analyze sales and profitability trends across months, states, and cities to identify high-performing and underperforming areas.

04 — Understand Customer & Payment Behavior

Explore repeat purchasing patterns and compare business performance across different payment methods.

05 — Generate Actionable Insights

Translate analytical findings into data-driven recommendations that support better profitability and business decisions.



Dataset & Data Preparation

Dataset Overview

The project is built on two related datasets: Orders and Details.

Orders Dataset

- 500 orders
- Order date
- Customer information
- State
- City

Details Dataset

- 1,500 transaction-detail records
- Sales amount
- Profit
- Quantity
- Category & Sub-category
- Payment method

Data Relationship

- The e-commerce business generates strong sales activity, but revenue growth does not consistently translate into healthy profitability.

Analysis Scope

- The dataset covers the full year of 2018, enabling analysis across products, customers, payment methods, regions, and time.

Data Preparation & Data Modeling

Data Preparation

Before analysis and dashboard development, the raw datasets were prepared to ensure consistency, accuracy, and usability.

Key Preparation Steps

Data Type Standardization

- Converted date fields to proper date format and standardized analytical fields.

Data Cleaning

- Removed formatting inconsistencies such as leading/trailing spaces and standardized categorical values.

Data Validation

- Checked for missing values, duplicates, unmatched order IDs, invalid quantities, and profit anomalies.

Data Modeling in Power BI

- The source data was transformed into a structured analytical model using multiple dimension tables connected through relevant keys, enabling efficient filtering, aggregation, and analysis.

Result

- A clean, structured, and analysis-ready data model was created for the Power BI dashboard.

Data Quality & Validation

Overall Data Quality

The datasets were found to be largely complete and consistent, with no missing values or duplicate records identified.

Validation Checks

- Missing Values: No missing values detected.
- Duplicate Records: No full duplicate rows identified.
- Order ID Matching: All 500 Order IDs were successfully matched across the two source datasets.
- Data Types: Date fields were standardized for analysis.
- Categorical Consistency: Formatting inconsistencies such as trailing spaces were corrected.

Data Anomalies

A small number of records showed Profit values greater than Amount.

These records were flagged for validation rather than modified, since the exact business definition of Amount and Profit needed to be preserved.

Final Outcome

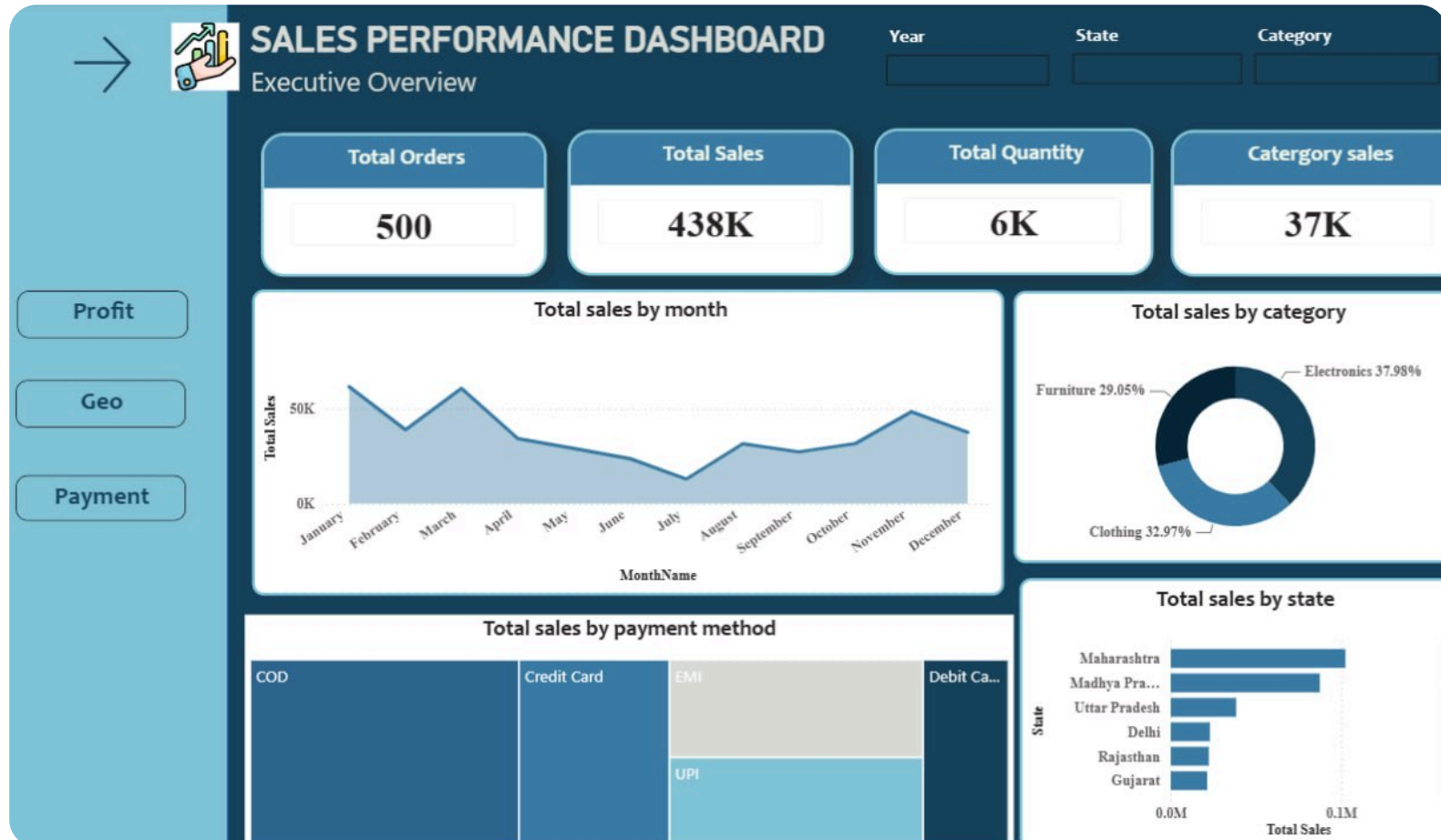
The resulting dataset was validated and prepared for reliable exploratory analysis and Power BI reporting.



Exploratory Data Analysis

Overall Performance

Strong sales activity, but profitability remains uneven across the business.



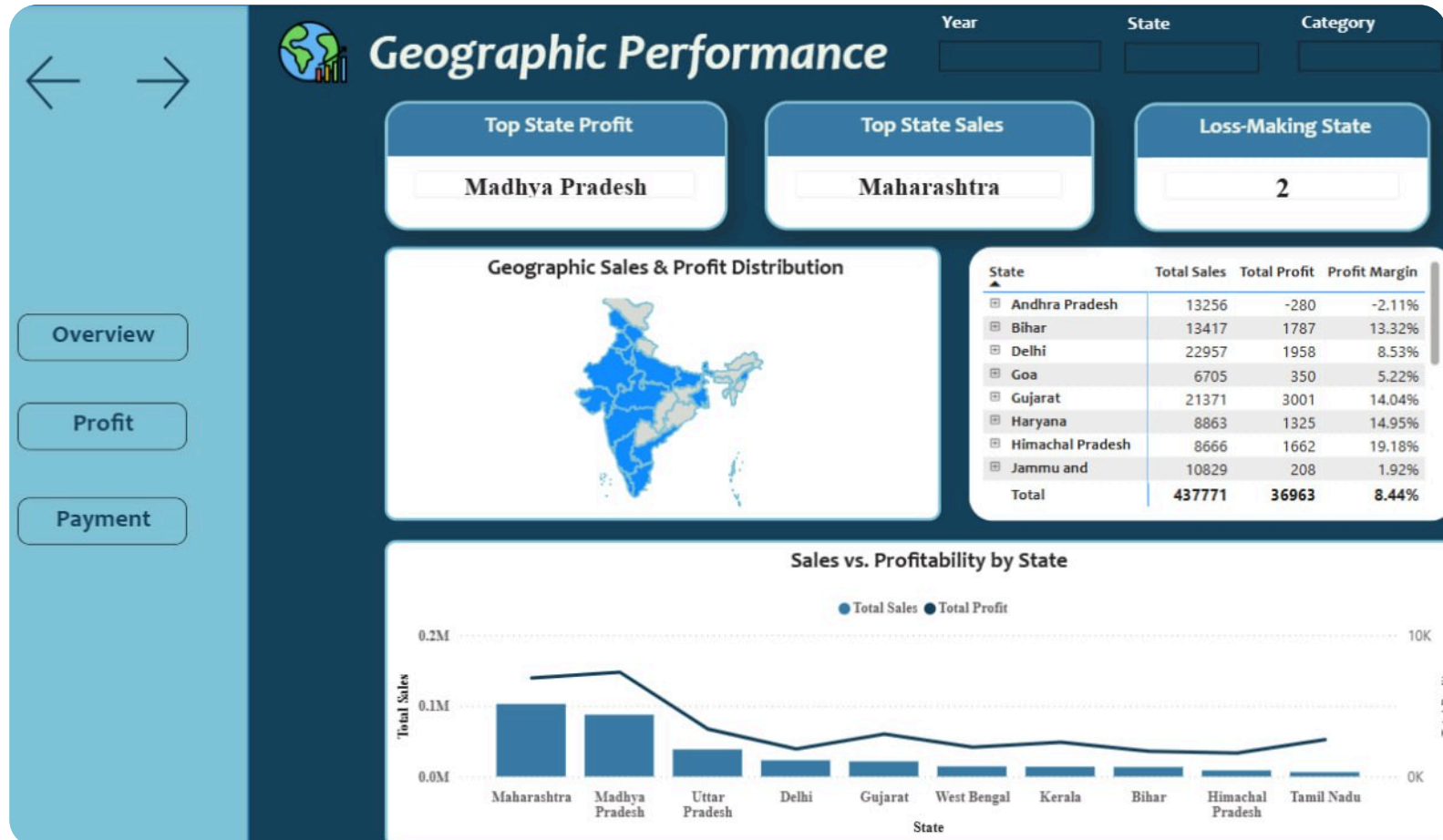
Product Performance

High sales do not always mean high profitability.



Geographic Performance

Regional performance varies, with some markets generating sales but negative profit.



Payment Analysis

Profitability differs across payment methods, despite differences in sales volume.





Key Findings & Insights

Key Findings

Sales ≠ Profit

Electronics generated the highest sales, but Clothing generated the highest profit.

Profit Leakage

33.8% of orders were loss-making.

High-Sales, Low-Profit Areas

Several sub-categories generated strong sales while producing negative profit.

Uneven Regional Performance

Some states generated meaningful sales but still reported negative profit.

What the Analysis Reveals

Revenue growth alone is not enough.

01 — Product Mix Matters

.Some high-sales categories and sub-categories contribute less to profitability

02 — Performance Varies Across Time & Markets

.Strong sales periods and regions do not always generate strong profit



03 — Profitability Should Drive Decisions

The focus should shift from maximizing sales volume to improving profitable growth.



Business Recommendations

Business Recommendations

01 — Optimize Loss-Making Products

Review pricing, costs, and promotional strategy for sub-categories generating negative profit.

02 — Investigate High-Sales / Low-Profit Segments

Identify why strong revenue is not translating into profitability.

03 — Focus on Profitable Markets

Investigate regions with meaningful sales but negative or weak profit.

04 — Learn from High-Performing Periods

Analyze the drivers behind strong-profit months and replicate successful practices.

05 — Monitor Profitability, Not Sales Alone

Use profit margin and loss-making orders alongside sales KPIs when making decisions.



Conclusion & Future Improvements

Conclusion & Future Improvements

From Sales Growth → to Profitable Growth

Conclusion:

The analysis shows that strong sales do not always translate into strong profitability. Identifying loss-making products, underperforming markets, and periods of weak profitability can help the business focus on more sustainable growth.

Future Improvements:

- Add more historical data to identify long-term trends and seasonality.
- Enrich customer-level data to support deeper retention and customer value analysis.
- Investigate cost and pricing drivers to explain the causes of negative profitability.
- Monitor key profitability KPIs continuously through the Power BI dashboard.

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